Suggested problems: 1, 3, 7, 8, 10

Problem 1: Prove that if $A \subset B$, then $B^c \subset A^c$.

Problem 2: Suppose that a number x is to be selected from the real line S, and let A and B be the following events:

$$A = \{x : 1 \leq x \leq 3\},$$
$$B = \{x : 3 \leq x \leq 7\}.$$

Describe each of the following events.

(a) A^c

(b) $A \cup B$

(c) $A^c \cap B^c$

Problem 3: Suppose that a coin is tossed three times. Identify an appropriate sample space for this experiment, and then identify each of the events as subsets of the sample space.

(a) The sample space S is...

(b) The event A that at least one head is obtained.

(c) The event B that a head is obtained on the second toss.

(d) The event C that a tail is obtained on the third toss.

(e) The event D that no heads are obtained.